

OPERATOR'S MANUAL

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CHAPTER EIGHT

THEORY OF OPERATION

Introduction

This chapter begins by presenting some of the underlying physics incorporated into the Infinity™ Nd:YAG laser system. Special attention will be given to those processes and features that are unique to the Infinity laser. It is, however, not necessary to read this chapter in order to operate the laser on a day to day basis. The chapter concludes with a review of the process of harmonic generation; specifically for the second, third and fourth harmonic configurations as they are employed in this laser system.

Laser Principles

By rearranging the configuration of their surrounding electrons, atoms and molecules can acquire energy from and give energy to their environment. Since the configurations the electrons can assume are limited to certain patterns according to the laws of quantum mechanics, the flow of energy is also limited to occur at certain discrete values in these processes. Following the absorption of an energy quantum the system (the word system will be used interchangeable for atoms and molecules) is said to be in an excited state. Since the system strives to return to its ground state (i.e., the state with the lowest possible energy) it will dispose of its excess energy. This can be done in several different ways, but if the system is left undisturbed it will do so by emitting one or more photons whose total energy will correspond to the energy difference between the excited state and the ground state. This process is called spontaneous emission and it occurs with a probability equal to the inverse of the lifetime of the excited state. In the literature this quantity is written as $A_{12} [s^{-1}]$, where the index refers to the energy levels involved. The process is illustrated in Figure 8-1a.

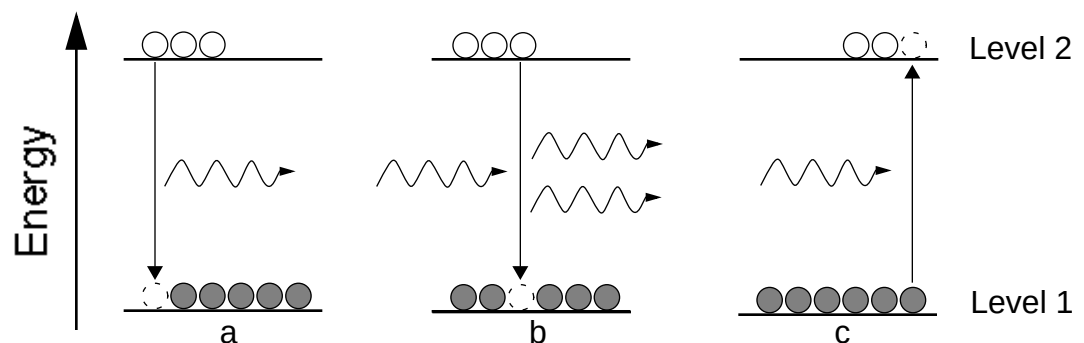


Figure 8-1. Illustration of Emission and Absorption Processes in Two Level System.
a. Spontaneous Emission, b. Stimulated Emission and c. Absorption.

The total rate of systems undergoing spontaneous de-excitation can now be written as $A_{21} N_2$, where N_2 is the number of systems in the excited state. If the excited system instead is disturbed by a radiation field in its vicinity, and the frequency of this external field is close to the resonance frequency of the excited system ($\omega = (E_1 - E_2) / 2\pi\hbar$), the excited system can be “stimulated” to return to the ground state as shown in Figure 8-1b. When this occurs a photon with identical properties (the same energy, direction and polarization) to the incoming one is emitted in the process. This process is called stimulated emission and it is the underlying mechanism of all laser action. The reason for this is that it provides a way of generating large numbers of identical photons in a controlled way (i.e., a coherent amplification mechanism). Stimulated emission occurs with a probability of B_{21} which is related to the A_{21} factor introduced in the previous section through the relationship: $B_{21} = A_{21} 2\pi c^3 / (\hbar \omega^3)$. Now, if there are N_2 numbers of excited systems, the de-excitation rate due to stimulated emission can be written as $B_{21} N_2 I$, where I [$\text{W}/\text{cm}^2 \text{ Hz}$] is the energy density in the external field. This process is illustrated in Figure 8-1b.

If the system instead is in the ground state when it is exposed to an external field oscillating at its resonance frequency, it can absorb an energy quantum and end up in an excited state through a process called stimulated absorption. This process is illustrated in Figure 8-1c. The two processes, stimulated absorption and emission are symmetrical with respect to each other. This means that an incoming photon does not distinguish between the upper and lower energy state and they hence occur with equal probability, $B_{21} = B_{12}$.

The three processes described above are called radiative since they involve the creation or annihilation of a photon. In addition to these processes there are also various non-radiative de-excitation and excitation mechanisms such as collisions with other systems and electron impact. Under normal conditions all of these processes are present simultaneously.

Since stimulated emission is the process in which laser light is generated we would like to arrange things so that this process dominates over the others. The only way to do this is to have a greater number of systems in the higher lying state so that $B_{21} N_2 > B_{12} N_1$, which is the same as the criteria $N_2 > N_1$. The situation when there are more systems in the higher lying state is called population inversion and it is a necessary condition for lasing action to take place. It is not possible to

achieve population inversion in a two level system of the type shown in Figure 8-1. The reason for this is that the faster systems are pumped into the excited state, the faster they will also undergo stimulated emission back to the ground state. Hence at least three energy levels are needed in a system where population inversion is to be created. A disadvantage of a three level system is that the lower level in the lasing transition will coincide with the ground state. This means that the population difference $N_2 - N_1$ will be smaller than if the lower level would have been an originally empty state. It is therefore advantageous to instead use a gain medium in which four levels are used, the situation illustrated in Figure 8-2.

The material used in the Infinity system, Nd:YAG, can be described by this four level scheme and the corresponding energy levels in this material and the spectroscopic nomenclature, are shown in Figure 8-2. Besides having the advantage of being a four level system, Nd:YAG also possesses a number of other attractive properties that make it suitable as a laser gain medium. After diamond and sapphire it is the hardest material known to man, its thermal conductivity is excellent and rods of high optical quality can be produced. Because of its narrow fluorescence line width it displays a low lasing threshold and high gain.

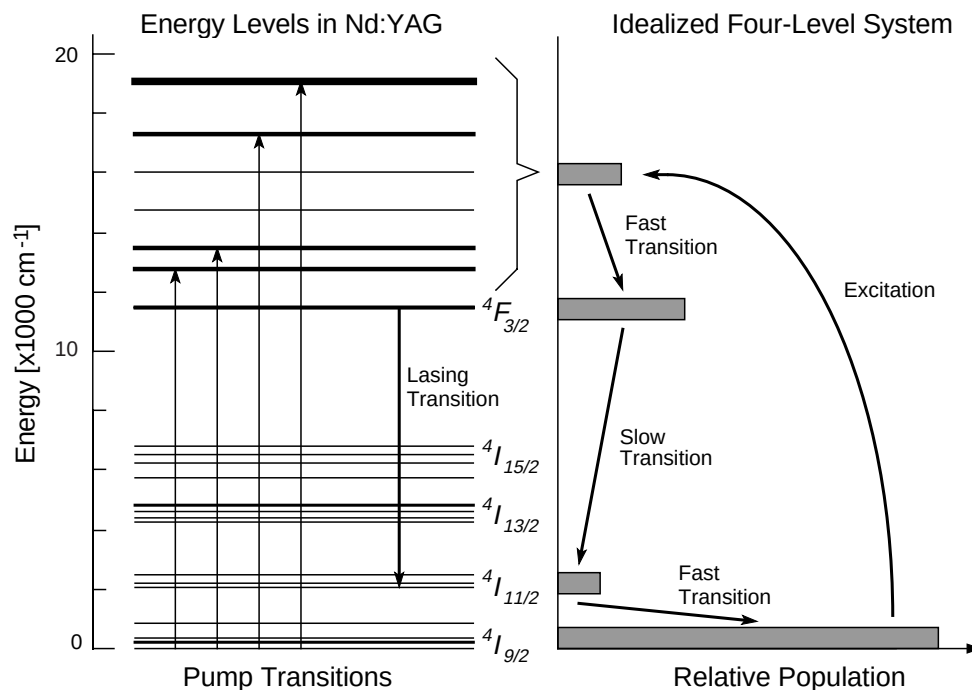


Figure 8-2. Energy Levels for a Nd:YAG System (Left) and a Four Level System (Right).

Laser Physics

Since the invention of the laser in 1960, it has been put to use in a great variety of applications. Despite the enormous multitude of laser types, they all share some common features. In order for a laser to produce coherent light a minimum of three conditions needs to be fulfilled. These three items are:

A Gain Medium

This is simply a medium in which energy from an external energy source, such as an electrical current or another laser, can be converted into a population inversion. If a population inversion is present in the medium then an incoming photon, having a frequency close to the corresponding energy difference, will with a greater probability be amplified through stimulated emission than absorbed through stimulated absorption. A gain medium is thus a coherent amplifier for photons.

A Pump Source

In order to prepare the system so that a state of population inversion is achieved, energy needs to be supplied. There are many different ways by which this can be done. The two most common are optical excitation, used in dye lasers, and electrical excitation (electron impact for example) used in gas discharge lasers like HeNe and Ar⁺ lasers. In the Infinity system the excitation energy is supplied by a flash lamp positioned close to the laser rods inside a highly diffusive cladding which ensures excellent optical coupling of the light from the flash lamp into the laser rods.

A Resonator

The function of the resonator is to provide the feedback necessary for the stimulated emission process and to confine the laser field within the boundaries of the gain medium. The simplest type of resonator consists of two plano parallel mirrors positioned with the gain medium between them as shown in Figure 8-3.

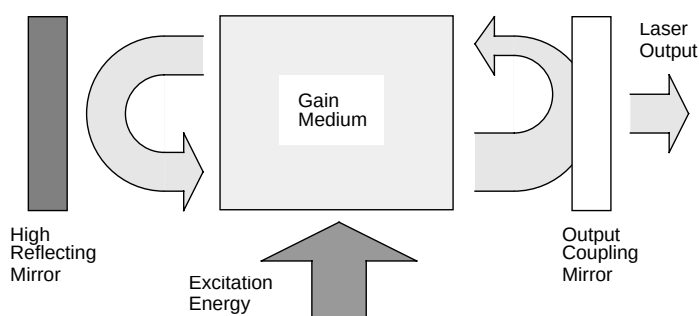


Figure 8-3. Illustration of Laser Resonator.

One of the mirrors has a reflectance of very close to 100% for the laser wavelength while the other allows some of the generated radiation to leak out of the cavity and hence has a lower reflectance, around 80% for example. The mirrors are called the high reflector and the output coupler, respectively. Some lasers have more complicated resonator arrangements incorporating more than two mirrors and/or other optical elements. The idea is still the same though, to confine the laser field inside the cavity, in a manner so that it reproduces itself after one cavity round-trip.

Even though an arrangement incorporating the three items described above is sufficient for producing laser emission other components are often added to it in order to control the polarization of the output, the spatial and temporal mode as well as the frequency width of the output. Once the pulse has left the oscillator its fundamental temporal and spatial properties are established. By passing it through one or more amplifying stages its brightness can, however, be increased (An amplifier is here simply a gain medium without a resonator). When doing so it is important to design the amplifier so that it does not introduce new or amplify present distortions. In Nd:YAG material this means that thermal effects need to be carefully controlled. Due to the nature of the phase conjugate mirror arrangement used in the Infinity laser, the distortions induced in the beam during the amplification process are kept at a minimum.

Laser Light Characteristic

Due to the nature of the process in which the laser light is generated, it has a number of unique properties which set it apart from conventional light sources. It is instructive to take a closer look at how the laser radiation compares to the light from an ordinary light source like an incandescent bulb. In comparison lasers are:

- **Brighter.** This means that the number of photons that a laser can provide per area and solid angle [$\text{W}/\text{cm}^2\text{sr}$] surpasses that of any other known source of light. This is an important parameter in optical systems since, using only passive components such as lenses, the brightness can not be increased outside the radiation source.
- **Coherent.** A coherent light source is one which maintains a fixed phase relationship between different points. Usually a distinction is made between spatial and temporal coherence. Spatial coherence is a measure of how well the phase can be predicted at different points in space at a given time while temporal coherence is a measure of how well the phase can be predicted at a

given point in space as a function of time. Coherence is hence a measure of the degree of phase correlation. A perfectly monochromatic light source would have infinite coherence length (time).

- nearly **Monochromatic**. The combination of narrow bandwidth and high peak power exhibited by lasers is one of their greatest assets. It allows the user to manipulate matter with an extreme degree of selectivity, and to perform advanced experiments in fields such as interferometry, holography and non-linear optics.
- **Tunable**. Using modern dye lasers, optical parametric oscillators, Raman shifting and various non-linear optical frequency conversion schemes, continuously tunable wavelengths from below 200 nm up into the infrared can be generated.
- **Directional**. This might be one of the laser's most striking characteristic. Laser beam propagation is limited only by diffraction. As a consequence laser beams can transport energy over enormous distances (e.g., to the moon and back) and be focused to very small spot sizes.

Even though the output from an ordinary light source, such as an incandescent lamp could be filtered to yield the same spatial and temporal coherence as any given laser source, the laser would still be more coherent. This is because the laser exhibits higher order coherency due to its different photon statistics. Photons from an ideal laser follow Poisson statistics while photons from a classical light source follow Bose-Einstein statistics. What this means is that the photons exiting from the laser can be viewed as statistically independent (i.e., detecting one photon does not aid in predicting when the next one will come). The filtering required to achieve the same temporal and spatial coherence as a laser would also result in an enormous loss of brightness.

Diode Pumped Lasers

Diode pumped solid state lasers are attractive for a number of reasons. They can be made very small, they convert electricity to laser light efficiently, they can be designed to lase in a single longitudinal mode and they are less sensitive to changes in their environment than flash lamp pumped systems. Furthermore the output from GaAs laser diodes, lasing in the region around 800 nm, have an excellent match with the main Nd:YAG absorption band (the $4F^{5/2}$ band). Flash lamps in comparison emit their light over a wide spectral range, covering the near ultraviolet, the visible and near infrared which means that most of the flash lamp output is converted into heat in the laser head. The fact that the output from laser diodes conveniently can be focused into the gain medium also contributes to reducing thermal effects in the rods. Because of the lower thermal load on the rods, problems with thermal lensing and thermally induced birefringence (thermally induced depolarization) are much less severe than in flash lamp pumped oscillators. As a result small diode pumped solid state lasers exhibiting excellent spatial and temporal behavior can be built. The life span of laser diodes, as measured in number of pulses fired before failure, is also significantly longer than that of flash lamps. State of the art flash lamps can fire up to several hundred million shots compared to more than one billion for laser diodes.

DPSS/DPMO

The Infinity master oscillator consists of two concave mirrors, a Nd:YAG prism and an acousto optical modulator arranged so that they form a ring resonator, see Figure 8-4.

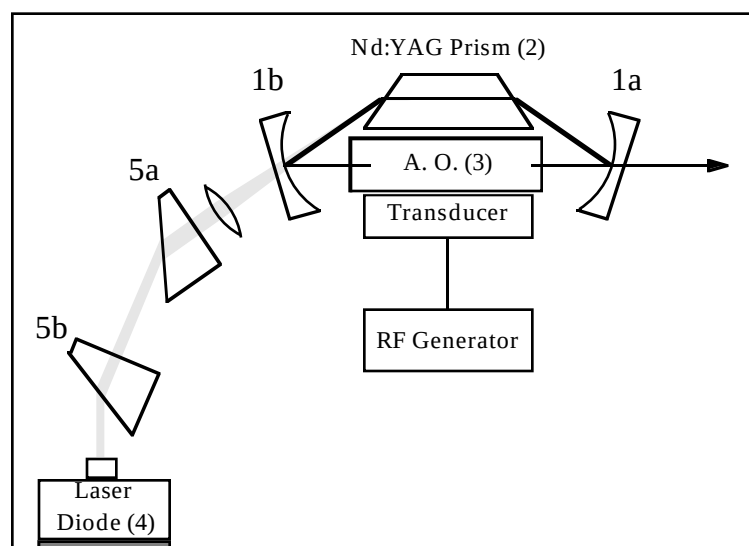


Figure 8-4. Infinity Master Oscillator Optical Layout.

In a ring formed cavity, instead of having a standing wave as in a linear cavity, two counter propagating waves can be present. Since these beams will leave the cavity in different directions, as shown in Figure 8-5, only one is usually useful. It is hence desirable to eliminate one of the beams so that all the stored inversion energy is made available for the direction used. This can be accomplished by introducing an additional loss mechanism for one of the directions. In the Infinity system the same device that is used to Q-switch the cavity, the acousto optical modulator, AOM, is used for this purpose.

Acousto-Optic Modulator

The acousto-optic modulator serves two purposes. It allows the round trip cavity losses to be regulated, providing the oscillator with a Q-switch. It also introduces a somewhat higher loss in the beam path for one of the two lasing directions. The last property, in combination with careful design of the oscillator geometry, ensures its unidirectional operation.

The underlying mechanism by which the modulator regulates the round trip losses is by diffraction of the optical wave against the acoustical wave induced in the material. As the modulator is switched on an acoustic wave is set up in the medium. The acoustic wave gives rise to a periodic variation in the medium's refractive index through the photoelastic effect. When the optical wave strikes this refractive index grating two new beams are created, one shifted up, the other down by the frequency of the acoustic wave. The directions of the new beams are given by conservation of momentum and energy in the photon - phonon interaction. Since the diffracted beams only can travel in a direction in which they undergo positive interference the optical path length difference between two adjacent rays must satisfy the following relationship

$$\lambda = 2\Lambda \sin(\Phi_B)$$

where λ is the wavelength of the optical radiation, Λ the wavelength of the acoustic wave and Φ_B the angle between them. Depending on if the incoming optical wave is scattered against or away from the acoustic wave the frequency of the diffracted beam will either be higher, $\omega_0 + \Delta\omega$, or lower, $\omega_0 - \Delta\omega$, than the incoming. These two processes are illustrated in Figure 8-5.

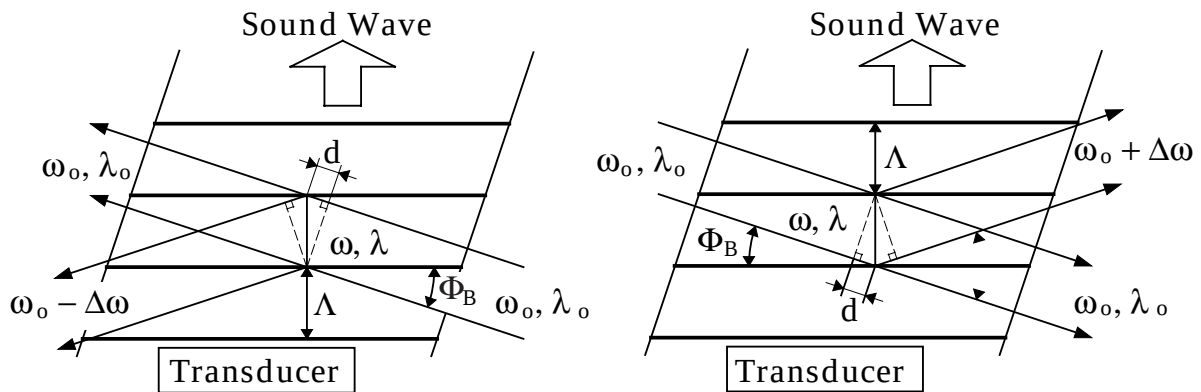


Figure 8-5. Geometry for Bragg Diffraction.

By switching on the acoustic field, energy is diffracted out of the main beam, introducing an additional loss mechanism in the cavity. When the transducer is momentarily switched off the acoustic wave decays, full transmission through the AOM crystal is restored and a laser pulse is created. A special feature of the design in the Infinity is that even when the transducer is on, the Q-switch is not fully closed but allows the oscillator to lase at a few milliwatts of CW power. This prelasing, which, as mentioned above, only occurs in one direction, provides the initial seed photons from which the next laser pulse builds up. This design enables the laser to be operated with exceptional stability with respect to energy, power, directionality and frequency.

Power Amplifier

By adding one or more amplifying stages after the master oscillator, a combination of high power, excellent beam quality and narrow band width can be achieved. The main function of the amplifier is to increase the brightness of the beam and to do so without introducing new or amplifying already present distortions.

Factors limiting the size of a Nd:YAG amplifier are a combination of crystal growth techniques and issues such as rod cooling and thermally induced stress and birefringence.

When designing an amplifier there is a trade-off between achieving high energy extraction or high gain. In order to extract most of the energy stored in the gain medium, the incoming pulse needs to already have an appreciable energy. If on the other hand high gain is desired then the incoming pulse should have an energy much smaller than the stored. In the case when the input energy is so small that the energy in the amplified pulse is still small compared to the stored, the amplification factor is called “small signal gain”. In the

opposite case, when the input energy is comparable to the stored energy the amplification factor will approach unity and the gain is said to be saturated.

In the Infinity system a multi-pass arrangement is used. This allows the amplifier to be used both as a small signal amplifier, exhibiting high gain but poor energy extraction (first pass) and as a low gain and high energy extraction efficiency amplifier (second pass).

An important consideration in high gain systems such as Nd:YAG amplifiers is to suppress the amplification of spontaneous emission (ASE) which will decrease the energy available for coherent emission and increase the incoherent background emission. ASE can also propagate backwards into the oscillator, affect its performance and potentially damage it. Methods that have been developed to deal with ASE are spatial filters, since ASE diverges more rapidly than laser emission, and Faraday rotators acting as optical diodes. In the Infinity system suppression of ASE is solved by a combination of careful design of the optical elements in the amplifier and by the use of a SBS mirror. Until threshold is reached its reflectance is essentially zero. As it is turned on by the incoming laser field, the energy stored in the laser rods is rapidly consumed by the coherent laser emission.

Polarization Control

A thermal gradient is created across the rod diameter as the radiation from the flash lamp is absorbed in it. The temperature gradient in turn gives rise to a stress gradient because the hotter regions in the center of the rod will try to expand more than the cooler surface regions. The resulting stress gradient gives rise to a refractive index gradient across the rod surface. Since the radial and the tangential stress components are different, the refractive indices along these two directions will be different as well. As a consequence the electro-magnetic vector components of the incoming radiation along the radial and the tangential direction, will experience different refractive indices and hence travel at different speeds through the rods.

The pulse exhibits horizontal polarization as it enters the first amplifying rod. Due to the difference in refractive index along the radial and tangential direction, the pulse exiting has acquired an elliptical polarization in all points across its spatial mode except for those that coincide with the two diameters that are parallel and perpendicular to the polarization state of the incoming pulse.

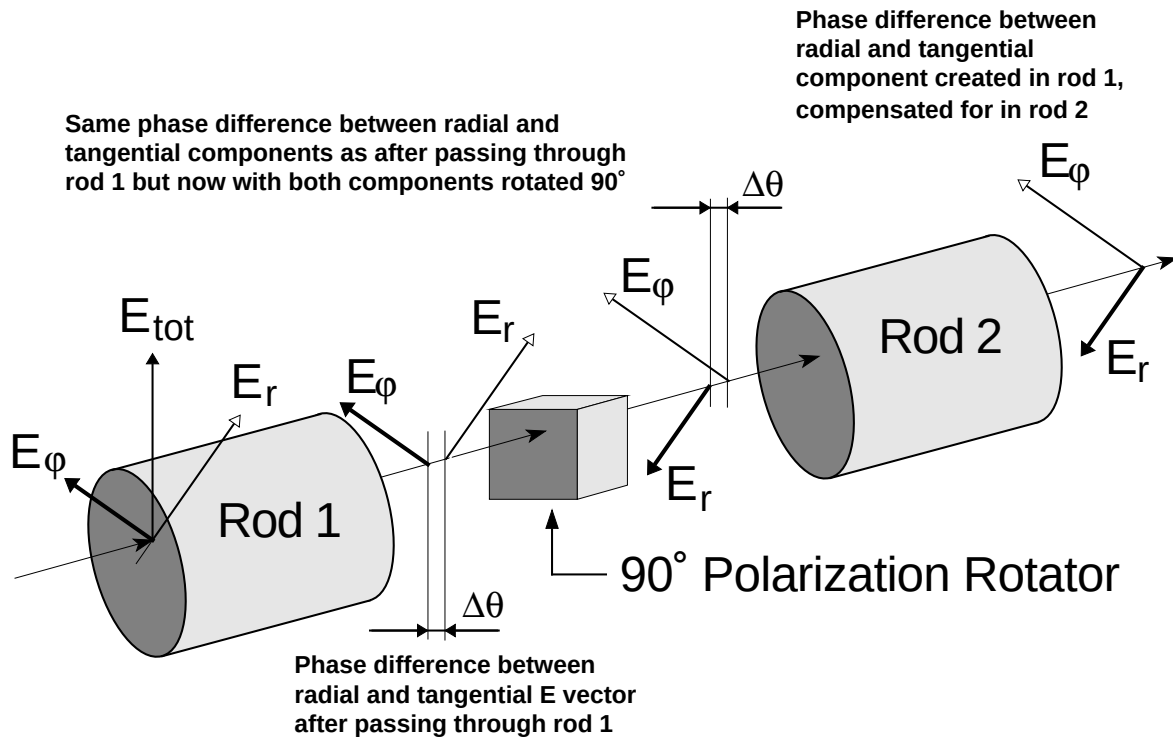


Figure 8-6. Polarization States of the Laser Pulse Before, Between and After Passing Through the Two Nd:YAG Rods.

If the pulse exiting the first rod, exhibiting elliptical polarization, is allowed to directly propagate through the second rod, the degree of ellipticity will increase, i.e. double. It is important to avoid this for the following reason.

After the pulse has been reflected in the SBS cell and passed through the two rods a second time the vertically polarized component of it will be reflected out of the laser amplifier by the thin-film polarizer. If the polarization state is poorly defined as the beam strikes the thin-film polarizer a large fraction of it will end up in the Faraday isolator. This energy is hence not available to the user and it can also, potentially, disturb the stability of the oscillator. It is therefore important that the beam is in a well defined polarization state as it goes through the quarter-wave plate, positioned between the second rod and the SBS cell.

To compensate for the phase difference induced by the birefringence in the first rod, a 90° polarization rotator is positioned between the two rods, see Figure 8-6. Light entering the first rod (from the left), having vertical polarization (horizontal in the laser, vertical used in the figure for illustration purpose only), will experience a phase

retardation between the radial, E_r , and tangential, E_t , component. Before the pulse enters the second rod the polarization is rotated 90° which means that the component that was radial in the first rod, now is the tangential and the tangential component in the first rod is now the radial. This means that the phase retardation that was induced in the first rod will be compensated by the second rod and the net result will be light that is again linearly polarized but now rotated 90° . The same correction will take place as the pulse goes through its second pass through the rods and the net result will be vertically linearly polarized light.

Relay Imaging

In order for the polarization control scheme described in the previous paragraph to work, the pulse traveling through the amplifier must experience the same relative environment in the two rods. Assuming that the two rods are close to identical and symmetrically loaded, this requires that the beam in the first rod is imaged into the second rod. Figure 8-7 illustrates how this is accomplished. The imaging, performed by two lenses and a concave mirror, is done so that the transverse coordinates remain unaltered while the coaxial coordinates are flipped around the center of the rods, see arrows inside rods in Figure 8-7. This arrangement not only performs part of the polarization control but also compensates for small residual asymmetries in the pumping of the rods. The relay cell is inserted in the beam path to avoid laser induced air breakdown at the focal points formed by the imaging system.

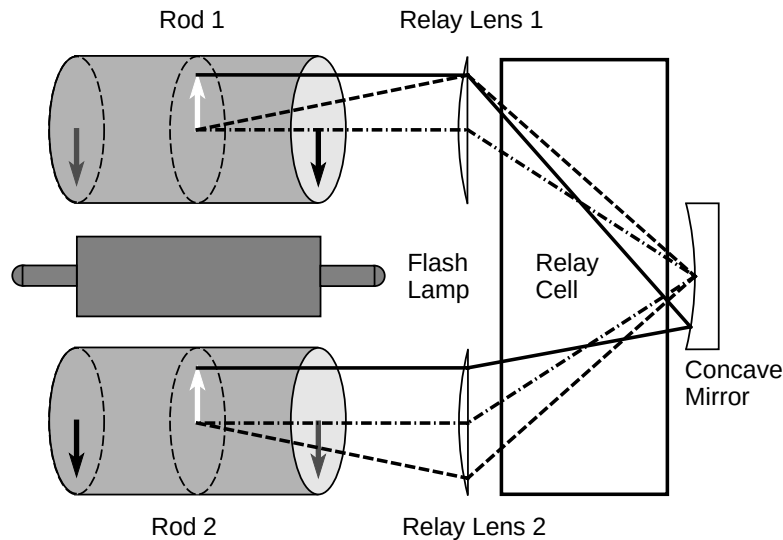


Figure 8-7. Principle of Relay Imaging.

Stimulated Brillouin Scattering

When light waves are scattered by density waves in a medium, a backwards traveling and frequency shifted component can be created in a process called Brillouin scattering. The density waves can be excited thermally or by the laser field itself through electrostriction (i.e., the macroscopic strain the electric field gives rise to). In the latter case the phenomenon is called Stimulated Brillouin Scattering or SBS. As is typical for processes that are self amplifying (stimulated), a minimum power density, a threshold, must be exceeded for it to occur. As long as the fluency is below this threshold value the process does not take place at all. When the fluency exceeds the threshold value the signal grows exponentially until limited by the onset of saturation. When the leading edge of the laser pulse returns through the rods it will decrease the gain. The effect of this can be seen in the trailing end of the laser pulse where there is a sudden drop in intensity.

A detailed theoretical analysis of the situation where a high power laser beam is focused into, for example CFC 113, shows that the field generated in the backwards direction by SBS is also the phase conjugate of the incident field. An illustration of how the reflectance of an incoming wave is different in a Phase Conjugate Mirror compared to an ordinary one is shown in Figure 8-8.

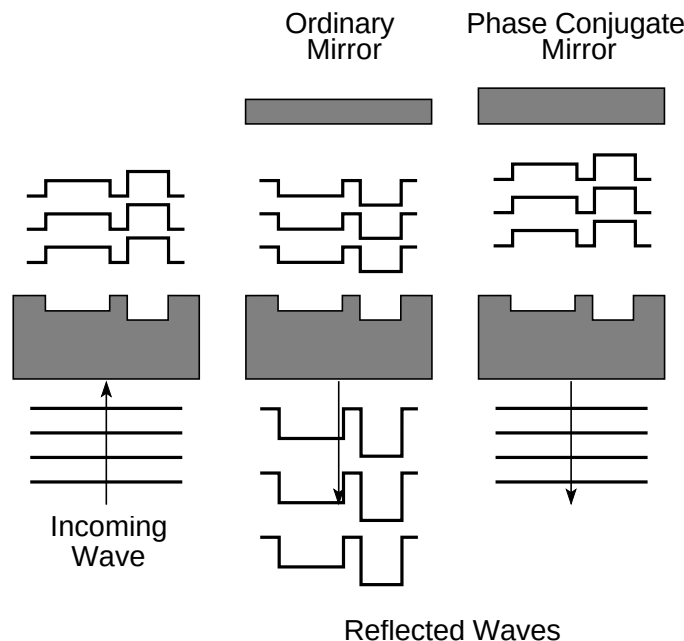


Figure 8-8. Illustration of the Difference Between Reflection in a Regular and a Phase Conjugate Mirror.

The pulse that undergoes reflection in the ordinary mirror, will after having traversed the distorting media a second time suffer from twice the distortion it had after its first pass while the pulse reflected in the phase conjugate mirror will be restored to its original shape.

In the Infinity laser this phenomena is used to correct for wave front distortions created by thermal and stress induced effects in the Nd:YAG rods. The distortions imposed on the pulse during its first pass through the amplifier are reversed on its second pass. Since this process is self correcting it is insensitive to changes in the degree and art of distortion. As a result the Infinity system can produce a laser pulse of superior quality that is independent of repetition rate and output power. There is, however, no polarization information in the phase conjugate process. This means that processes resulting in changes in the polarization state of the beam are not reversed or corrected. Instead the polarization state of the laser pulse is determined by the oscillator and the design of the beam path.

Harmonic Generation

The primary radiation from the Infinity laser is emitted at 1064 nm. For most applications, an output wavelength in the visible is, however, desired. This can be achieved by doubling the frequency of the Nd:YAG fundamental in a non-linear optical process called Harmonic Generation. A brief description of this process is given below.

Applying an electrical field to a linear dielectric medium induces a polarization in it according to the relationship:

$$P = \epsilon \chi E, \quad (1)$$

where E is the electric field vector, ϵ the permittivity of free space and χ is a dimensionless constant (electrical susceptibility) characteristic for the polarized material. A medium which can be described by this relationship is said to be linear. For a long time it was believed that all materials were linear and this assumption was one of the cornerstones of optics. Consequences of this assumption are that:

- optical properties are independent of light intensity
- the frequency of light does not change and
- light does not interact with light.

With the advent of the laser it has, however, become possible to create E-fields large enough to saturate the polarization. As a consequence the induced polarization will no longer be related to the electrical field strength in a linear way. For such cases a more general expression must be used. It is convenient to write this as a series expansion in the electrical field according to:

$$P = \epsilon (\chi_1 E + \chi_2 E^2 + \chi_3 E^3 + \dots), \quad (2)$$

where χ_1 is the regular linear susceptibility and χ_2, χ_3 are the second and third order non-linear susceptibilities respectively. Normally $\chi_1 \gg \chi_2 \gg \chi_3$ and hence the non-linear effects will be negligible unless the E-field is very large. If an applied field can be described by

$$E = E_0 \sin(\omega t) \quad (3)$$

the induced polarization, according to Eq. 2, takes the form:

$$P = \epsilon \chi_1 E \sin(\omega t) + \epsilon \chi_2 E^2 \sin^2(\omega t) + \dots \quad (4)$$

which can be rewritten, using standard trigonometric relationships, as:

$$P = \epsilon \chi_1 E \sin(\omega t) + 1/2 \epsilon \chi_2 E^2 \{1 - \cos(2\omega t)\} + \dots \quad (5)$$

In the last expression, the presence of the second term shows that a wave having twice the frequency of the fundamental can be formed in a non-linear medium. This phenomena is called frequency doubling or Second Harmonic Generation (SHG).

Since the refractive index is wavelength dependent it will usually be different for the fundamental and the frequency doubled light. This means that over a distance of several wavelengths the net frequency doubled power will be zero. In order to generate any substantial amount of frequency doubled light, the two refractive indexes needs to be the equal so that the phase difference between the waves remain constant. This situation is called phase matching and it can be achieved in materials that are birefringent (materials which have different refractive indexes for different polarization states). Figure 8-9 shows a cross section of what is called a phase-matching ellipsoid. The circles and ellipses show how the refractive indices, for two different polarization states and wavelengths, varies as a function of angle to the optical axis, which in the figure coincides with the Z-axis. As can be seen there is an angle, Θ_m , for which the refractive index for the fundamental beam, n_o^ω , is equal to the refractive index of the frequency doubled beam $n_e^{2\omega}$. When the crystal is tilted so that the incoming radiation propagates along this line it is said to be phase matched and frequency doubling will take place with the highest possible efficiency in the non-linear material.

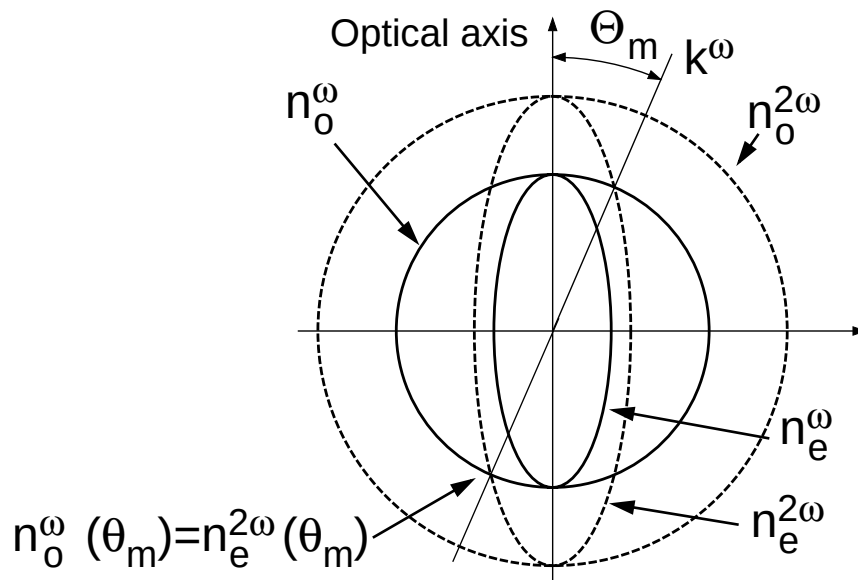


Figure 8-9. Illustration of Phase Matching in a Birefringent Media.

Since n_o and n_e have different temperature derivatives, phase matching can also be achieved by tuning the crystal temperature. In the Infinity system the crystals are, however, angle tuned.

After the crystal the two wavelengths can be separated using either a prism or, which is more common, a dichroic mirror that transmits one wavelength and reflects the other.

There are two different polarization arrangements that can be used for SHG, called type 1 and type 2 doubling. In the Infinity laser system type 1 doubling is used because it is the more efficient process of the two in BBO.

For some applications such as pumping of Optical Parametric Oscillators and of dye lasers running on Coumarin dyes, shorter wavelengths than 532 nm are desirable. Through a non-linear optical process similar to the one described above, radiation at 355 nm and at 266 nm can be produced starting with the Nd:YAG fundamental at 1064 nm. Third Harmonic Generation (THG) of 355 nm radiation is accomplished by frequency mixing, in a second crystal, the fundamental 1064 nm radiation of the Infinity laser and the frequency doubled radiation. Since energy is conserved, the resulting wavelength is given by $1 / (1 / 1064 + 1 / 532)$ which is equal to 355 nm. Radiation at 266 nm is produced by removing the left over 1064 nm after second harmonic generation, and then frequency doubling the 532 nm beam.

Second Harmonic Generation

After the pulse has been reflected out of the amplifying system by the thin film polarizer, it is sent through an angle tuned BBO crystal (assuming that this option has been installed). The crystal is mounted in a cell enclosed by two antireflection coated windows. The entrance and exit surfaces of the crystal have also been antireflection coated. In the crystal, the infrared radiation undergoes type I doubling which means that the polarization of the fundamental remains unchanged while the polarization of the second harmonic at 532 nm is perpendicular to the incoming fundamental, as shown in Figure 8-10. A dichroic mirror is used to separate the fundamental wavelength from the second harmonic.

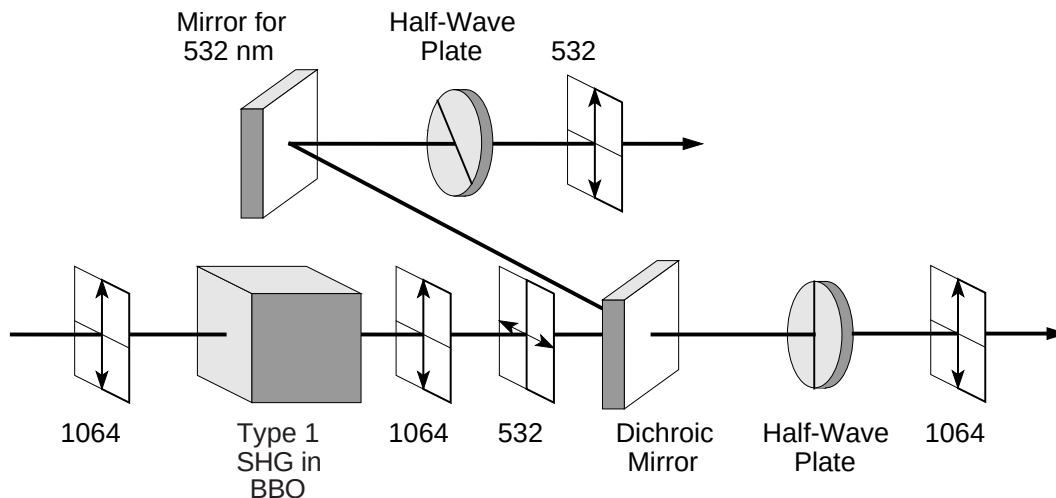


Figure 8-10. Polarization States Before and After Second Harmonic Generation.
(The Half-Wave Plates are Optional in the SHG Options Package).

Third Harmonic Generation

If the optional Third Harmonic Generation package has been installed, the beams (1064 nm and 532 nm) exiting the SHG crystal go through a wave plate, acting as a full wave plate for the second harmonic and as a half wave plate for the fundamental wavelength. The beams are then sent through a second angle tuned BBO crystal where they undergo Type I frequency mixing. The radiation produced at 355 nm is thereafter separated from the infrared and doubled radiation in a dichroic mirror, Figure 2-11.

Provisions are also made for installing an additional (and optional) dichroic mirror in the beam path containing the 532 and 1064 nm radiation. This dichroic (sometimes referred to as a stripper) directs the 1064 nm radiation into a beam dump while the 532 nm radiation is transmitted.

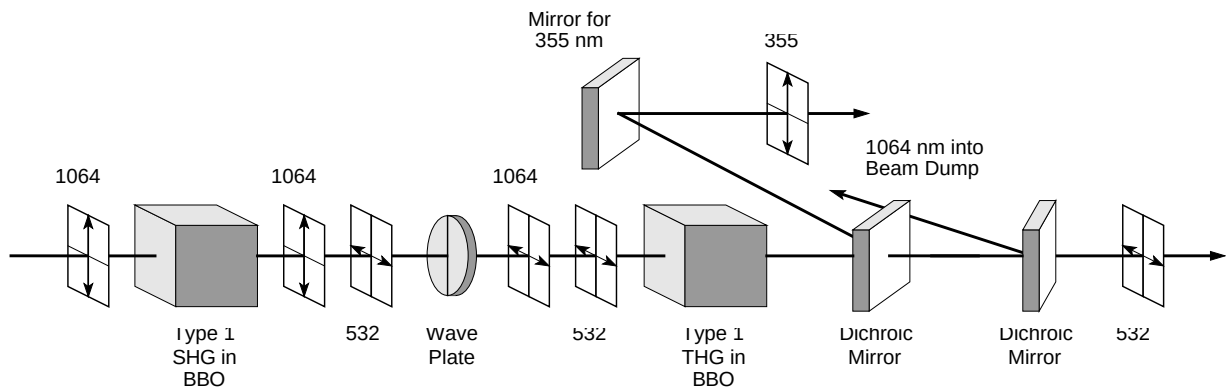


Figure 8-11. Polarization States Before, During and After Third Harmonic Generation.
(Dichroic Separating 1064 nm from 532 nm is Optional in THG Package Option)

Fourth Harmonic Generation

Radiation at 266 nm can be produced by frequency doubling the light at 532 nm, produced in the SHG process. This is done by adding one more BBO crystal to the set-up used for SHG. Since the IR fundamental is not used again after the SHG generation it is removed by a dichroic mirror, see Figure 2-12 (this dichroic is sometimes referred to as a stripper). After fourth harmonic generation the 532 nm and the 266 nm beams are separated in a dichroic mirror.

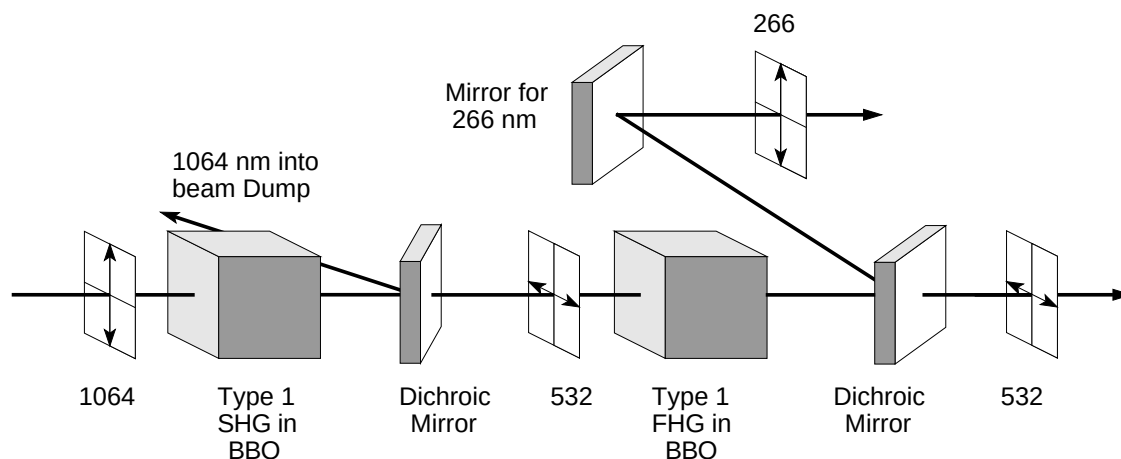


Figure 8-12. Polarization States Before, During and After Fourth Harmonic Generation.